

Quantitative Trading & Strategy Architecture

Comprehensive Technical Handbook & Educational Directory

Domain	Primary Focus	Core Methodologies	Target Assets
Quantitative Finance	Alpha Model & Strategy Building	Machine Learning, AR(1), Stacking, Microstructure	Crypto (BTC, ETH), FX, Equities

EXECUTIVE SUMMARY: This technical manual provides a comprehensive, source-grounded breakdown of quantitative trading as taught across the MemLabs video library by a former high-frequency trading (HFT) quant trader. It covers mathematical foundations, feature engineering, statistical edge formulation, machine learning architectures (PyTorch, XGBoost, Model Stacking, Adaptive/Online Learning), order book microstructure, and end-to-end execution strategy design.

1. Core Foundations & Mathematical Framework

Defining Quantitative Trading: Quant trading is the formal process of engineering a **statistical edge** (a positive Expected Value, $EV > 0$) using mathematical models, and executing it systematically to generate **risk-adjusted returns**. Execution is equally as critical as the edge itself; poor order execution or improper trade sizing can completely eradicate a valid statistical edge.

Concept / Metric	Mathematical Formulation	Quantitative Meaning & Operational Role
Expected Value (EV)	$EV = E[X] = \text{SUM}(P(\text{win}) * \text{WinSize} - P(\text{loss}) * \text{LossSize})$	Measures average net P&L per trade. Strategy is viable only when $EV > 0$, even with low win rates (e.g. 51%).
Sharpe Ratio (Annualized)	$\text{Sharpe} = (E[r] / \text{StdDev}(r)) * \text{sqrt}(N_periods)$	Quantifies return per unit of risk. Higher Sharpe smooths equity curves, minimizes drawdowns, and permits higher safe leverage.
Log Return (Symmetric)	$r_t = \ln(P_t / P_{t-1})$	Provides time-additivity (summing log returns models compound growth) and mathematical symmetry (up and down moves match).
AR(1) Linear Model	$y_{\text{hat}}_{t+1} = w * y_t + b$	Simple 2-parameter model: $w < 0$ indicates Mean Reversion (pullback to mean), $w > 0$ indicates Momentum (directional persistence).

The Law of Large Numbers & High Trade Frequency: Institutional quantitative funds (such as Renaissance Technologies' Medallion Fund) rely on tiny statistical edges—winning as little as 51% to 53% of trades with an EV of 8 basis points (0.08%). While 10 trades yield negligible profit, executing thousands of trades per day allows the sampled EV to converge rapidly via the Law of Large Numbers, compounding modest edge into exceptional, stable returns.

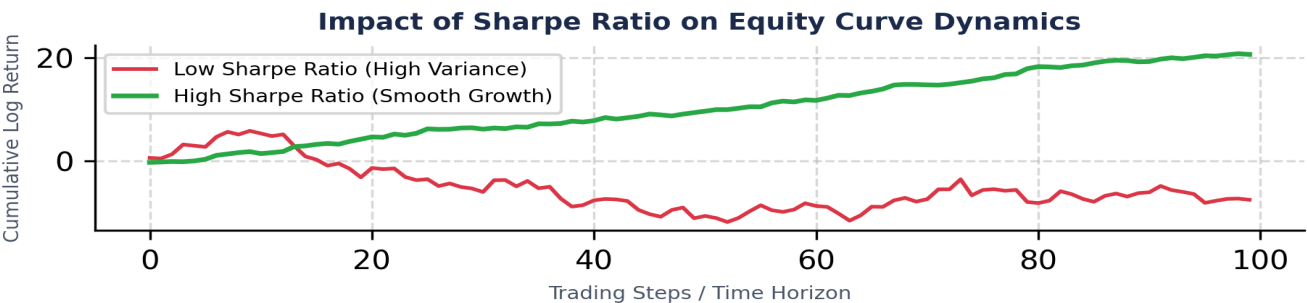


Figure 1.1: Visualizing Sharpe Ratio impact on cumulative P&L stability and drawdown control over time.

2. Market Microstructure & Feature Engineering

Order Book Microstructure: At the core of financial markets is the order book, containing limit bids (buy commitments) and limit asks (sell commitments). The **mid-price** $((Best\ Bid + Best\ Ask) / 2)$ provides a clean signal for model inputs, eliminating the *bid-ask bounce* noise present in raw trade prices. Analyzing order flow imbalance—large market orders crossing multiple book levels relative to resting liquidity—allows quant traders to forecast short-term mean reversion.

Feature Category	Engineering Calculation	Quantitative Signal & Utility
Lagged Log Returns	$r_{\{t-k\}} = \ln(P_{\{t-k\}} / P_{\{t-k-1\}})$	Extracts short-term auto-regressive memory; captures mean reversion or momentum across discrete lags.
Directional Encoding	$dir_{\{t-1\}} = \text{sign}(r_{\{t-1\}}) \text{ in } \{-1, 0, +1\}$	Isolates price direction from magnitude noise; prevents extreme outlier returns from distorting discrete probability states.
Rolling Window Mean	$MA_n(r) = (1/n) * \sum_{\{i=1\}^n} r_{\{t-i\}}$	Compresses past returns into a scalar hidden state; captures broader regime memory while eliminating forward data leakage via shifting.
Order Book Imbalance	$Imbalance = \ln(Buy_Volume / Ask_Liquidity)$	Quantifies pressure crossing the book. High buying volume against thin liquidity predicts imminent mean reversion.

Preventing Data Leakage: A critical failure mode in quantitative backtesting is forward data leakage. When calculating rolling statistics or lag features, functions must explicitly apply a *shift(1)* (e.g., in pandas or Polars) before rolling computations. This guarantees that feature vectors at time *t* contain strictly historical information up to *t-1*.

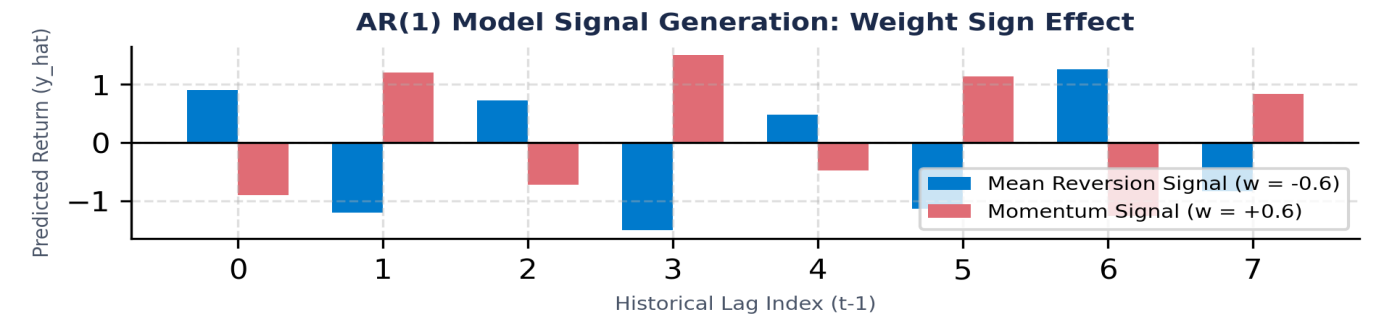


Figure 2.1: Directional signal response of AR(1) models given negative weight (Mean Reversion) vs positive weight (Momentum).

3. Machine Learning Architectures & Model Stacking

Quantitative strategies leverage diverse ML paradigms based on structural requirements, interpretability, and data non-stationarity:

Model Paradigm	Primary Implementation	Key Strengths & Mechanism	Non-Stationarity Handling
Univariate Linear (AR1)	PyTorch / SGD / Closed-Form	High interpretability (2 parameters: w, b). Computes in 3-5 CPU cycles. Generalizes well against noise.	Requires rolling feature memory or re-training splits.
Non-Linear Tree (XGBoost)	XGBoost Classifier / Regressor	Captures piece-wise non-linear expectations (e.g. return conditioned on weekday). Constrained max_depth=3 prevents overfitting.	Requires time-based splits (75% train / 25% test) to verify out-of-sample persistence.
Online Learning	PassiveAggressive Regressor	Streams data row-by-row. Dynamically adjusts weights on error without re-training full batches.	Inherent adaptation to distribution shifts and sudden market regime changes.
Reinforcement Learning	Reinforce Policy Gradient	Models trading as a 2-armed bandit. Uses Entropy Regularization to prevent getting trapped in local optima.	Dynamically shifts action policy (betting long vs short) when environment reward structure flips.
Model Stacking	Constrained Linear Meta-Learner	Combines predictions of non-linear base models (e.g., Weekday XGB + Moving Average XGB) via positive weights (w >= 0) and bias.	Produces more robust, higher Sharpe signals than any individual standalone model.

EMPIRICAL MODEL STACKING RESULTS: Model Stacking (Meta-Learning) Superiority: In empirical backtests on ETH/BTC data, individual models yielded expected trade returns of 40 bps (Weekday XGBoost) and 24 bps (30-day Directional XGBoost). By feeding their predictions into a constrained linear meta-learner with positive weights ($w \geq 0$) and a 17 bps negative bias, the stacked ensemble achieved an **expected return of 47.6 bps** and a **259% gross return** without experiencing negative equity across out-of-sample test periods.

4. Strategy Execution, Trade Sizing & Risk Management

Market Making vs. Market Taking: Quantitative execution strategies broadly split into two taxonomies:

- **Market Taking:** Uses market orders to remove liquidity. Subject to higher taker fees. To remain profitable, taking strategies require longer forecast horizons (e.g. 8–12 hours) so that expected trade returns exceed roundtrip transaction costs.
- **Market Making:** Posts limit orders to provide liquidity, capturing the bid-ask spread and receiving maker rebates. Market makers dynamically skew/bias their bid and ask quotes around the mid-price based on model predictions.

Sizing Strategy	Mathematical Formula	Operational Dynamics & Risk Profile
Constant Sizing	$\text{SignedSize} = \text{Sign}(\hat{y}) * \text{Capital}$	Fixed dollar allocation per trade. Baseline method, does not compound gains or scale with signal strength.
Piecewise Linear	$\text{SignedSize} = \text{MaxSize} * \text{HardTanh}(\hat{y})$	Scales trade size linearly with predicted return magnitude $\hat{y}$, strictly clipped at upper and lower bounds.
Smooth Non-Linear	$\text{SignedSize} = \text{MaxSize} * \tanh(\hat{y})$	Gradual hyperbolic tangent scaling. Highly sensitive to subtle signal shifts while naturally bounding maximum exposure.
Compounding Sizing	$\text{EntryValue}_t = \text{Capital} * \exp(\sum r_i)$	Reinvests accumulated P&L continuously using log return additivity. Magnifies long-term returns when paired with positive EV.

Leverage & Liquidation Risk: Leverage (L) scales both profits and losses ($L * P\&L$). Excessive leverage lowers the liquidation threshold $P_{\{liq\}}$ when prices breach maintenance margin limits. Leverage must be calibrated strictly against the strategy's Sharpe ratio and worst-case drawdown.

5. Statistical Validation & Walk-Forward Testing

Validation Technique	Structure & Implementation	Key Objective / Risk Mitigation
In-Sample / Out-of-Sample	75% oldest data for training, 25% most recent for testing.	Primitive check for signal persistence without temporal shuffling.
Expanding Window CV	Training set grows iteratively (t_0 to t_k), testing on fixed future step $t_{\{k+1\}}$.	Simulates historical walk-forward accumulation of market data.
Rolling Window CV	Fixed-size window slides forward in time ($t_{\{i\}}$ to $t_{\{i+n\}}$), dropping oldest data.	Tests model adaptability strictly under recent market regimes.
Girko Excess Predictability	Compares model return against a uniform random baseline strategy.	Quantifies true excess predictive power independent of general market trend.

SYSTEMIC TAKEAWAY: The Quant Trading Equation: $\text{Alpha} = \text{Statistical Edge} * \text{Execution Quality}$. A profitable trading system combines a high-Sharpe, cross-validated model (Mean Reversion, XGBoost, or Stacked Ensemble) with robust execution mechanics (time-based or quote-biased, log-compounded sizing, and disciplined leverage management).